

Grade 6 Accelerated
Unit 14
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

1. A great white pelican can hold about 700 cubic inches in its bill. After flying down to try to catch a fish, the pelican has 100 cubic inches of water in his mouth. The inequality $700 \geq 100 + w$ represents this situation, with w representing the amount of space that can still be filled in the pelican's bill. What is the maximum amount of water that can be added to the bill without spilling out?

$$\begin{array}{r} 700 \geq 100 + w \\ -100 \quad -100 \\ \hline 600 \geq w \end{array} \quad \text{Less than or equal to 600 in}^3$$

2. Clay County is gearing up for their annual Math Field Day competition. They need to order boxed lunches for all the proctors. Chic-Fil-A charges them \$6.75 per box for lunch. The budget for the event lunches is \$225. The inequality $6.75x \leq \$225$ represents the situation, with x representing the number of box lunches that can be purchased from Chic-Fil-A. What is the maximum number of boxed lunches that can be purchased?

$$\begin{array}{r} 6.75x \leq 225 \\ \hline 6.75 \quad 6.75 \\ \hline x \leq 33.\bar{3} \end{array} \quad \text{33 boxed lunches}$$

3. The band department is raising money to purchase new uniforms by selling Mums for senior night. The team earns \$0.75 for each flower sold. They hope to raise at least \$250 from this fundraiser. This situation is represented by the inequality $0.75x \geq \$250$. What is the minimum number of flowers that the band must sell to reach their fundraising goal?

$$\begin{array}{r} 0.75x \geq 250 \\ \hline 0.75 \quad 0.75 \\ \hline x \geq 333.\bar{3} \end{array} \quad \text{334 flowers}$$

4. Maximum capacity at Busch Gardens, Tampa is 33,000 people. The inequality $b \leq 33,000$ represents this situation, with b representing the number of patrons at the park. Give an example of a solution to the inequality that is not valid in this context. **Answers may vary.**

-1; a negative amount of people can't visit the park.

5. Hilary is buying cupcakes from Publix to share with her 26 classmates for her birthday. The store has 18 vanilla cupcakes on the shelf. She wants to buy the remaining chocolate for the remaining cupcakes. The inequality $18 + c \geq 26$ represents the situation, with c standing for the number of cupcakes that she still needs to buy. Give an example of a solution to the inequality that is not valid in this context.

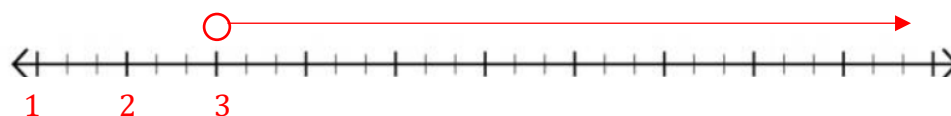
$$\begin{array}{r} 18 + c \geq 26 \\ -18 \quad -18 \\ \hline c \geq 8 \end{array} \quad \text{9.5 is not a valid because you can't buy a fraction or part of a cupcake.}$$

6. Find the solution to both inequalities using the division property of inequality.

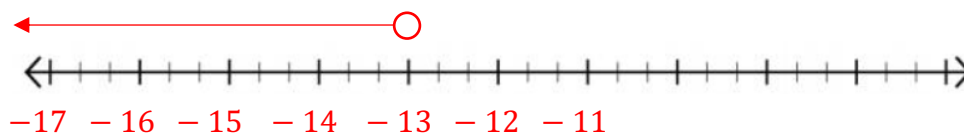
$\frac{7y}{7} < \frac{-105}{7}$ $y < -15$	$\frac{-7y}{-7} < \frac{-105}{-7}$ $y > 15$
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Solve each inequality. Then, graph its solution set.

1. $-2 \cdot -\frac{c}{2} < -1.5 \cdot -2$ $c > 3$



2. $\frac{-4x}{-4} > \frac{52}{-4}$ $x < -13$



3. $\frac{5x}{5} \leq \frac{-55}{5}$ $x \leq -11$



10. How is the process of solving an inequality with a negative coefficient similar to the process of solving an inequality without a negative coefficient?

In both situations, you use the Division property of equality and divide both sides by the negative coefficient.